



Faculty of Management Technology

Department of **Business Informatics**

The German University in Cairo

Artificial Intelligence in Fixed Asset Management

by:

Hassan Ahmed Eltayeb

Bachelor Thesis

Student ID: 58-16514

Date: 17th of May 2026

Supervisor: Dr. Ayman Alserafi

Abstract

The application of artificial intelligence (AI) technology in fixed asset management has become increasingly important for organizations. Fixed Asset Management (FAM) involves the control, monitoring, maintenance, depreciation, and accounting of fixed assets throughout their useful life. With the growing need for accurate asset valuation and better maintenance planning, there is a need for intelligent systems that can support decision-making in this area. This thesis focuses on the use of machine learning in fixed asset management, especially through a prototype system for depreciation prediction. The prototype uses asset-related inputs such as cost, age, usage level, maintenance information, and asset type to support outputs such as predicted asset value, depreciation amount, and estimated maintenance cost. The system is represented using prototype screens, BPMN models, a use case diagram, user stories, test cases, field requirements, and business rules. The thesis follows the Design Science Research Methodology (DSRM), where the artifact is designed, demonstrated, evaluated, and improved based on expert feedback. The evaluation was conducted with one technical expert and one accounting expert. Their feedback led to improvements in the output screen, prediction BPMN, use case diagram, user stories, and test cases. Therefore, the thesis shows that a machine learning-based prototype can support fixed asset management

Contents

1	Introduction	1
2	Literature Review	1
2.1	Artificial Intelligence	1
2.1.1	Definition of AI	2
2.1.2	Modern AI Architectures	2
2.1.3	Types of AI Tools	2
2.2	Fixed Asset Management	3
2.2.1	Types of Asset Management	3
2.2.2	Processes of Fixed Asset Management in Different Industries	3
2.2.3	Tasks in Fixed Asset Management	4
2.3	Artificial Intelligence in Fixed Asset Management	4
2.3.1	Technical Components	4
2.3.1.1	RAGs & LoRA	5
2.3.1.2	NLP	5
2.3.1.3	Computer Vision	5
2.3.1.4	Agentic AI	5
2.3.1.5	Machine Learning	6
2.3.2	Integration of AI Tools in Fixed Asset Management	6
2.3.3	Challenges of AI Implementation in Fixed Asset Management	6

2.4	Research Gap	7
3	Application:	9
3.1	Use Case Diagram	9
3.2	Business Process	10
3.2.1	Depreciation Prediction BPMN	10
3.2.2	Report BPMN	11
3.3	User Stories	12
3.4	Field Requirements and Business Rules	14
3.5	Wireframes	16
3.5.1	Model Input Screen	17
3.5.2	Model Output Screen	19
3.6	Test Cases	21
4	Research Methodology	23
4.1	Design science DSRM	23
4.1.1	Identify problem & motivate	24
4.1.2	Define objectives of a solution	24
4.1.3	Design & Development	24
4.1.4	Demonstration	24
4.1.5	Evaluation	25
4.1.6	Communication	25
5	Results	26
5.1	Application of Design Science Research Methodology (DSRM) . . .	26
5.1.1	Identify problem & motivate	27
5.1.2	Define objectives of a solution	27

5.1.3	Design & Development	27
5.1.4	Demonstration	28
5.1.5	Evaluation	28
5.1.6	Communication	37
6	Discussion	38
7	Conclusion and Future Work	40
Appendix A	Prototype Screenshots, Figma Links, and Prompts	45
A.1	Asset Depreciation Prediction Input Screen Prompt and Figma Link	45
A.2	Depreciation Prediction Results Output Screen Prompt and Figma Link	47
A.3	Improved Depreciation Prediction Results Output Screen Prompt and Figma Link	50
Appendix B	Interview Questions	53
B.1	Technical Expert Interview Questions	53
B.2	Accounting Expert Interview Questions	54
Appendix C	Interview Responses	55
C.1	Technical Expert Interview Responses	55
C.2	Accounting Expert Interview Responses	57
	Declaration	60

List of Figures

3.1	Use Case Diagram	10
3.2	Depreciation Prediction BPMN	11
3.3	Reporting BPMN	12
3.4	Asset Depreciation Prediction Input Screen	18
3.5	Depreciation Prediction Results Output Screen	20
5.1	Improved Depreciation Prediction Results Output Screen	34
5.2	Improved Depreciation Prediction BPMN	35
5.3	Improved Use Case Diagram	36

List of Tables

3.1	User Stories Table	12
3.2	Field Requirements and Business Rules Table	14
3.3	Test Cases Table	21
5.1	Improved User Stories Table	30
5.2	Improved Test Cases Table	32

List of Abbreviations

AI	Artificial Intelligence
FAM	Fixed Asset Management
LORA	Low-Rank Adaptation
ML	Machine Learning
NLP	Natural Language Processing
RAGs	Retrieval-Augmented Generations
RPA	Robotic Process Automation

Chapter 1

Introduction

Fixed Asset Management is an important aspect in organizational operations, since fixed assets, which include buildings, machinery, cars, and equipment among others, form a large portion of their total asset base. These fixed assets should be monitored and maintained throughout their life span until retirement. Gavrikova, Volkova, and Burda (2020) says that asset management is important for ensuring efficient utilization of the assets during their entire lifespan, where as Paul, Ogugua, and Eyo-Udo (2024) says that management of fixed assets comprises processes such as acquisition, authentication, installation, maintenance, and disposal of assets. Poor management of fixed assets will surely result in poor valuation and poor maintenance of the assets.

One important part of fixed asset management is depreciation. Depreciation helps organizations estimate how the value of an asset decreases over time. Traditional depreciation methods, such as the straight-line method, are useful because they are simple and easy to apply. However, these methods may not always reflect the

real condition of an asset because they usually depend on limited factors such as cost, useful life, and time. In reality, asset value can also be affected by other factors such as usage level, maintenance condition, and asset type.

Artificial Intelligence and Machine Learning can support fixed asset management by helping organizations use more asset-related information when estimating asset value and depreciation. In this thesis, the focus is on using a machine learning concept, specifically a regression-based approach, to support depreciation prediction. Regression is suitable for this type of problem because the system is expected to predict numerical outputs such as asset value, depreciation amount, and maintenance cost.

The aim of this thesis is to design a prototype system that shows how machine learning can be used to support fixed asset depreciation prediction. The prototype allows users to enter asset-related data such as asset cost, age, usage level, maintenance information, and asset type. Based on these inputs, the system presents predicted asset value, depreciation amount, and estimated maintenance cost. The prototype also includes supporting elements such as BPMN diagrams, a use case diagram, user stories, test cases, field requirements, and business rules.

This thesis is divided into several chapters. Chapter 2 presents the literature review, where previous studies related to artificial intelligence, machine learning, fixed asset management, AI tools used in asset-related processes, and the challenges of applying AI in fixed asset management are discussed. Chapter 3 presents the application part of the thesis, including the prototype wireframes, BPMN diagrams, use case diagram, user stories, test cases, and field requirements and business rules. Chapter 4 explains the research methodology, mainly the Design Science Research Methodology. Chapter 5 presents the results of applying the

methodology, including the expert evaluation and the improved prototype components. Chapter 6 discusses the results, the feedback received, the changes made to the prototype, and the lessons learned. Finally, Chapter 7 presents the conclusion, limitations, and future work.

Chapter 2

Literature Review

This literature review tackles three main sections. First, it discusses Artificial Intelligence, including the definition of AI, modern AI architectures, and the types of AI tools. Then, it covers Fixed Asset Management with its subsections on the types, processes, and tasks of fixed asset management. Lastly, it discusses the integration of the first two sections, namely AI in Fixed Asset Management, focusing on the technical components needed, the AI tools required for integration, and the challenges of such integration. At the end, it presents a BPMN model, a use-case diagram, and user stories.

2.1 Artificial Intelligence

Artificial intelligence is a broad field that supports intelligent decision-making, automation, and data-driven organizational processes. In this section, we discuss an overview of AI, its main architectures, and the major types of AI tools used in organizations.

2.1.1 Definition of AI

According to Jiang, Li, Luo, Yin, and Kaynak (2022), AI can be defined as the ability of systems to behave intelligently and perform activities that often require human intelligence. Similarly, Bughin et al. (2017) describe AI as an important technological improvement that is becoming increasingly recognized as the next digital frontier for businesses.

2.1.2 Modern AI Architectures

Jiang et al. (2022) explain that deep learning systems can operate through multiple computational layers that extract hierarchical patterns from data, allowing models to learn without direct human intervention. Similarly, Bughin et al. (2017) highlight that modern AI architectures help integrate machine learning with advanced analytics, forming the basis for automated decision-making in organizational settings. Davenport (2018) also note that AI includes autonomous agents capable of adapting their behavior based on continuous data inputs. Meanwhile, Kitsios and Kamariotou (2021) draw attention to hybrid architectures that combine internal knowledge bases with computational models, thereby linking organizational knowledge with learning algorithms.

2.1.3 Types of AI Tools

Huang and Rust (2018) classify AI tools into four categories: mechanical, intuitive, analytical, and empathetic systems. These categories reflect the variety of activities that AI tools can replicate within organizational processes. Kogan, Kokina, Stampone, and Boyle (2024) describe robotic process automation (RPA) as a tool that reduces human error and strengthens internal control environments by carry-

ing out rule-based tasks accurately. In addition, Hou, Yu, Lu, Yang, et al. (2017) describe computer vision as a tool that can interpret visual data for inspections, monitoring, and defect detection. Furthermore, Santos et al. (2023) show that natural language processing extracts information from unstructured documents and supports automation in text-based workflows, which is crucial in organizational document processing.

2.2 Fixed Asset Management

This section discusses the types of asset management, the processes of fixed asset management across different industries, and the tasks involved in managing fixed assets.

2.2.1 Types of Asset Management

Gavrikova et al. (2020) explain that asset management consists of tactical, strategic, and operational actions intended to ensure that assets continue to perform and provide value throughout their life cycle. Likewise, Ford et al. (2022) state that asset management practice operates on three levels: strategic, tactical, and operational, each aimed at aligning organizational objectives with asset performance.

2.2.2 Processes of Fixed Asset Management in Different Industries

Paul et al. (2024) state that fixed asset management depends on a series of systematic procedures, including acquisition, verification, utilization, maintenance, and disposal across organizational environments. Additionally, Hou et al. (2017)

note that the maintenance process involves identifying irregularities in equipment operation and anticipating possible faults before they occur, which is especially important in manufacturing and industrial environments.

2.2.3 Tasks in Fixed Asset Management

Paul et al. (2024) explain that fixed asset management tasks include acquisition, assigning asset identifiers, conducting physical audits, preparing depreciation schedules, and maintaining documentation throughout the asset life cycle. Likewise, Gavrikova et al. (2020) emphasize the importance of accurate information flows, continuous monitoring, and structured reporting.

2.3 Artificial Intelligence in Fixed Asset Management

This section discusses AI in fixed asset management, including the technical components needed to integrate AI into fixed asset management and the challenges of doing so. It also covers the accounting requirements analysis through a BPMN model, use-case diagram, and user stories.

2.3.1 Technical Components

This section outlines technical components used in AI, including Machine learning, RAGs, LoRA, natural language processing (NLP), computer vision, and agentic AI. It also discusses the integration of these technical components into FAM and the challenges that can occur during implementation.

2.3.1.1 RAGs & LoRA

Retrieval-augmented generation (RAG) and low-rank adaptation (LoRA) are important techniques that can support AI systems by improving knowledge access and model adaptation. In the context of fixed asset management, these techniques can help systems retrieve relevant records and adapt to organizational data more efficiently.

2.3.1.2 NLP

Santos et al. (2023) state that NLP extracts information from unstructured documents and enables automation in enterprise asset management processes. Similarly, Huang and Rust (2018) identify NLP as useful in interpreting maintenance logs, inspection notes, and service reports within asset management environments.

2.3.1.3 Computer Vision

Hou et al. (2017) explain that computer vision systems analyze visual data to support inspections and monitoring, making them highly relevant in asset management. Additionally, Ocumpaugh, Roscoe, Baker, Hutt, and Aguilar (2024) note that automated visual analysis supports continuous monitoring of physical environments while reducing the need for manual observations.

2.3.1.4 Agentic AI

Davenport (2018) show that AI systems can operate as autonomous agents that make recommendations, execute tasks, and adjust to real-time data. Likewise, Chen (2025) explain that agentic AI introduces autonomous systems capable of making decisions, analyzing data, and executing actions independently.

2.3.1.5 Machine Learning

Tulebayev, Mukhanova, and Goranin (2025) show that machine learning can support asset management by using variables such as asset age, depreciation level, and usage intensity to predict asset risk. Similarly, their study supports the idea that asset-related data can be used to improve maintenance decisions and provide more informed fixed asset management.

2.3.2 Integration of AI Tools in Fixed Asset Management

Fu and Li (2022) state that continuous monitoring supported by advanced learning models shows how AI contributes to better asset oversight. Moreover, Paul et al. (2024) indicate that tracking and maintenance functions become more efficient when supported by digital innovations, demonstrating that AI-related mechanisms enhance the accuracy of fixed asset processes. Additionally, Santos et al. (2023) explain that information extraction from unstructured documents through NLP improves documentation quality and reduces the time required to process asset-related records.

2.3.3 Challenges of AI Implementation in Fixed Asset Management

Wamba-Taguimdje, Fosso Wamba, Kala Kamdjoug, and Tchatchouang Wanko (2020) report that challenges in integrating AI into organizational environments include limited data quality and resistance to technological change. Furthermore, Mogaji and Nguyen (2022) identify trust and transparency as major challenges when integrating AI into compliance-driven sectors such as FAM.

2.4 Research Gap

Recent research shows that there are still several gaps in the use of artificial intelligence in fixed asset management. Paul et al. (2024) explain that many recent technological developments in fixed asset management focus mainly on asset tracking and maintenance automation. However, these studies do not focus enough on how artificial intelligence can support accounting-related tasks such as depreciation estimation and asset value evaluation.

Similarly, Santos et al. (2023) show that unstructured maintenance and service documentation can affect asset management processes. This shows that asset-related data is often difficult to use in a clear and structured way. In the context of this thesis, this is important because depreciation prediction depends on having useful asset information such as cost, age, usage, maintenance, and asset type.

In addition, Kogan et al. (2024) explain that automation technologies require clear processes, control mechanisms, and a strong understanding of business operations. However, limited research discusses how AI-based tools can be designed in a way that supports accounting users, managers, and auditors in fixed asset management. This creates a need for prototype-based studies that show how AI concepts can be represented through screens, BPMN diagrams, use case diagrams, user stories, test cases, and business rules.

Based on the reviewed literature, several research gaps can be identified. First, there is limited research on the use of artificial intelligence for accounting-specific fixed asset tasks, especially depreciation prediction and asset value estimation. Second, there is a lack of prototype-based studies that show how machine learning concepts can be translated into practical system designs for fixed asset manage-

ment. In addition, limited attention has been given to connecting technical system design with accounting requirements such as reporting, audit relevance, validation rules, and business interpretation of results. Finally, there is still insufficient discussion on how expert feedback from both technical and accounting perspectives can improve the design of AI-based fixed asset management systems.

How can the use of machine learning methods assist in improving the effectiveness of fixed asset management practices from a financial perspective such as the prediction of depreciation, valuation of assets, and interpreting maintenance costs?

Chapter 3

Application:

This chapter presents the application part of the thesis by showing the main design components of the proposed prototype, including the use case diagram, BPMN diagrams, user stories, field requirements and business rules, wireframes, and test cases.

3.1 Use Case Diagram

The use case diagram shown in Figure 3.1 provides an overview of the major interactions between the users and the proposed system for predicting depreciation. This figure specifies the primary roles involved in the system, such as the accountant and the manager, as well as the major functions available to them. These functions include adding asset information, creating predictions, displaying prediction results, reviewing summaries of input data, and producing reports.

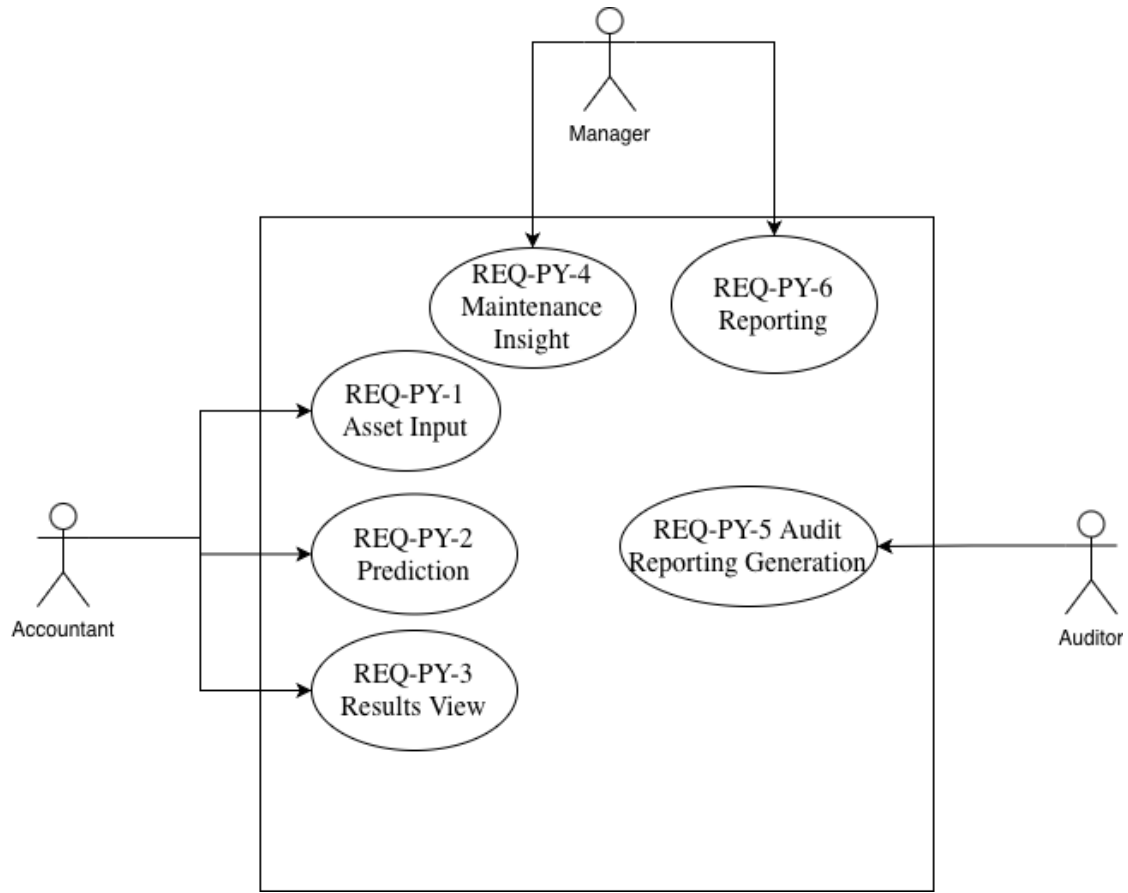


Figure 3.1: Use Case Diagram

3.2 Business Process

This section presents the business processes of the proposed prototype, including the depreciation prediction process and the report generation process.

3.2.1 Depreciation Prediction BPMN

Figure 3.2 shows how the business process works in the depreciation prediction system. The process starts when the user provides data such as the cost, age, usage level, maintenance cost, and asset category of the asset in question. After

entering the data, the system performs validation to verify that the data complies with the set business rules. If there is an error with the input data, the system gives an error notification to the user. When the data is validated, the processing phase is initiated using automatic service activities that depict the machine learning algorithm. This phase processes the input data and produces predictions on the depreciation rate and maintenance cost.

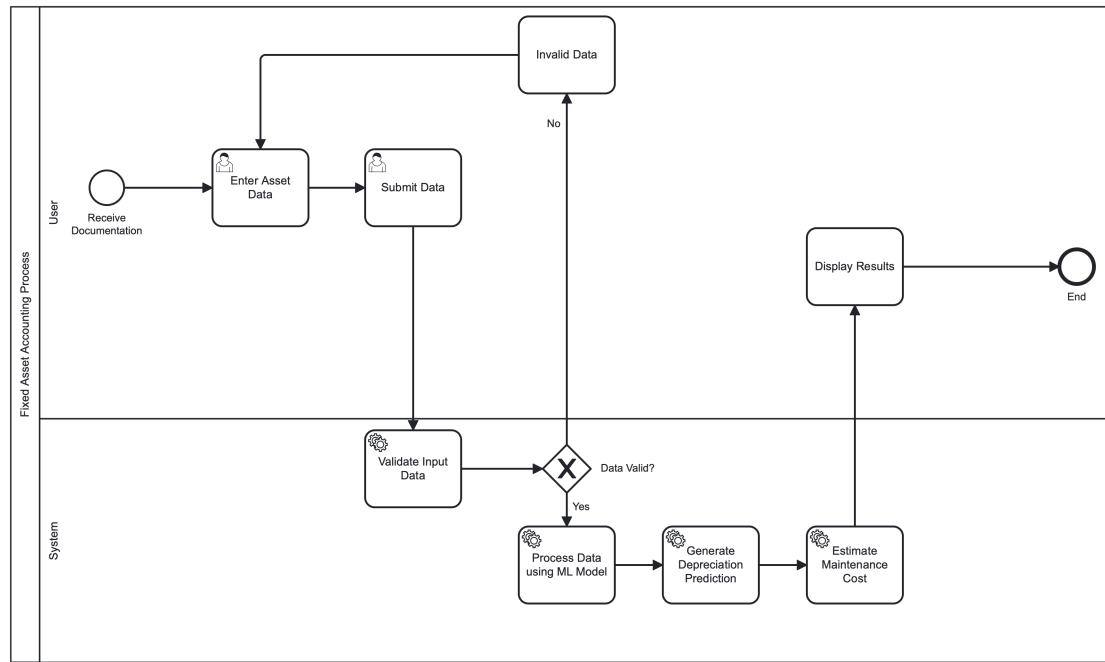


Figure 3.2: Depreciation Prediction BPMN

3.2.2 Report BPMN

Figure 3.3 demonstrates the procedure for report generation in the recommended system. Report generation starts when the user makes a report generation request based on the predictions made. The system retrieves the predictions and prepares the report. After preparing the report, it is delivered to the user. With this

procedure, users are able to access reports of the predictions made.

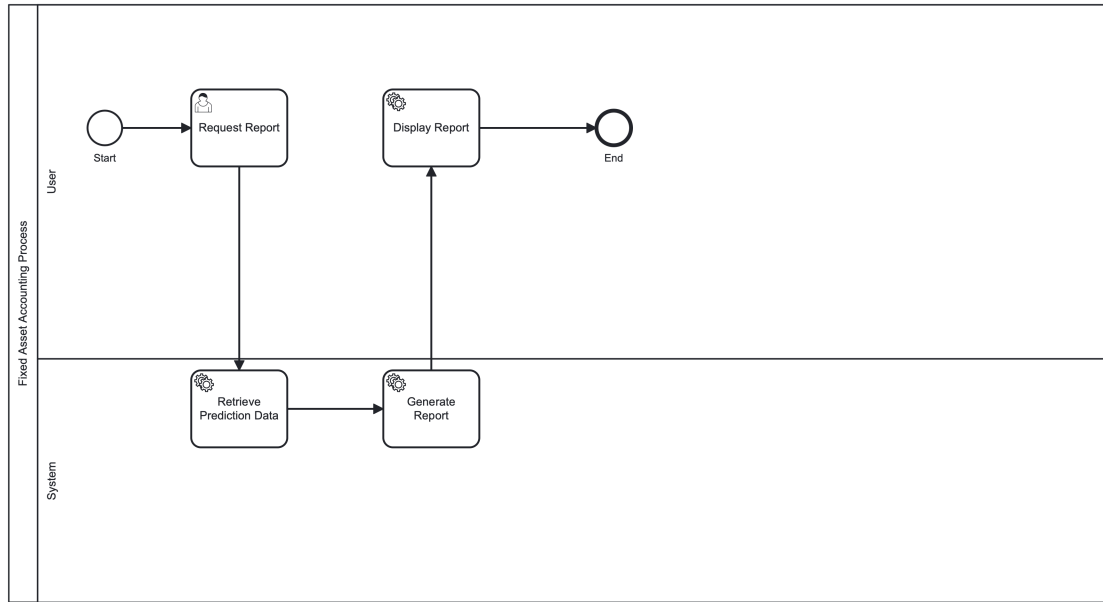


Figure 3.3: Reporting BPMN

3.3 User Stories

Table 3.1 presents the user stories of the proposed system, including the requirement ID, epic, and user story summary for each main system function.

Table 3.1: User Stories Table

Requirement ID	Epic	User Story Summary

REQ-PY-1	Asset Input	As an Accountant, I want to enter asset data (cost, age, usage, maintenance, asset type) so that the system can process it for prediction.
REQ-PY-2	Prediction	As an Accountant, I want the system to generate predicted asset value and depreciation so that I can analyze asset performance.
REQ-PY-3	Results View	As an Accountant, I want to view the predicted results.
REQ-PY-4	Maintenance Insight	As a Manager, I want to see estimated maintenance cost so that I can evaluate asset efficiency.
REQ-PY-5	Report generation	As an Auditor, I want to generate a report so that I can use the results for auditing.
REQ-PY-6	Reporting	As a Manager, I want to generate a report so that I can use the results for decision-making.

3.4 Field Requirements and Business Rules

Table 3.2 presents the field requirements and business rules that control the input data, output values, validation logic, and reporting behavior of the prototype.

Table 3.2: Field Requirements and Business Rules Table

Field / Output	Requirement Type	Requirement / Business Rule
Asset Cost	Field Requirement	The asset cost must be entered as a number greater than zero. The system should not accept zero or negative asset cost.
Asset Age	Field Requirement	The asset age must be entered as a number greater than or equal to zero. Negative age values are not accepted.
Usage Level	Field Requirement	The usage level must be selected from predefined options such as low, medium, or high in order to keep the input standardized.
Maintenance Information	Field Requirement	The maintenance field must be completed before the prediction process. It can represent the maintenance condition or maintenance cost depending on the prototype design.

Asset Type	Field Requirement	The asset type must be selected from predefined categories such as equipment, machinery, vehicle, or building. This helps the system process asset data in a consistent format.
Mandatory Fields	Business Rule	All required fields must be completed before the user can run the depreciation prediction process. Missing mandatory fields should lead to a validation message.
Predicted Asset Value	Business Rule	The predicted asset value must not be negative and should not be higher than the original asset cost. This keeps the output within a logical accounting range.
Depreciation Amount	Business Rule	The depreciation amount must not be negative and must not exceed the original asset cost. This prevents illogical depreciation results.
Estimated Maintenance Cost	Business Rule	The estimated maintenance cost must be displayed as zero or a positive value. It should not appear as a negative amount.

System Feed-back	Business Rule	The system should provide feedback when the entered data is valid or invalid, so the user can understand whether the prediction can be processed.
Asset Condition Insight	Business Rule	The asset condition insight should explain the meaning of the prediction results in a simple way, such as whether the asset is moderately depreciated or has average operational efficiency.
Report Generation	Business Rule	A report should only be generated after prediction results are available. The report should depend on the generated depreciation and maintenance outputs.

3.5 Wireframes

This section presents the prototype wireframes, including the model input screen and the model output screen used to demonstrate how users interact with the proposed system.

3.5.1 Model Input Screen

Figure 3.4 shows the data that the system is going to use to calculate the cost of an asset's depreciation and determine its current worth. In order to provide this data, users have to fill out certain pieces of data regarding the asset, which include cost, age, usage, maintenance, and type. Before sending data to the processing step, all data should comply with the business rules related to the validity of data. Cost, for instance, needs to be a positive number, and the age needs to be either zero or a positive number, all mandatory fields need to be completed before the start of the prediction. Usage and types are restricted to certain predetermined categories. In this case, all data passed to the processing step of the system will have a standardized format of representing an asset therefore leading to standardized output.

Asset Depreciation Prediction

 **Asset Cost**

 **Asset Age (years)**

 **Usage Level**

 **Maintenance Cost**

 **Asset Type**

Predict Depreciation

The system uses a machine learning model to estimate asset value and depreciation.

Figure 3.4: Asset Depreciation Prediction Input Screen

3.5.2 Model Output Screen

As shown in Figure 3.5, the screen provides an overview of the outcomes generated by the system after performing calculations on the data provided. To calculate these results, the system uses a regression-type machine learning model to process the inputs. The screen reveals the forecasted value of the asset, the amount of its depreciation, and the maintenance cost. In order to facilitate comprehension of all this information, the screen includes an overview of input data and system messages that indicate whether or not the entered data is within acceptable boundaries. There is also a section that provides insight into asset condition. All these outcomes are consistent with logical and business requirements. For example, the forecasted value cannot be greater than the original value of the asset or less than zero.

Depreciation Prediction Results

Review and adjust the ML-generated asset predictions

\$

Predicted Asset Value

\$ 7,500

↘

Depreciation Amount

\$ 2,500

🔗

Estimated Maintenance Cost

\$ 1,200

ⓘ

How the Prediction Works

Model Type

Regression-based Machine Learning Model

Inputs Used

Cost, Age, Usage, Maintenance, Asset Type

The model analyzes asset characteristics and historical patterns to estimate current value, depreciation, and expected maintenance cost. It uses advanced regression techniques to identify relationships between asset properties and their financial outcomes.

← Edit Input

Generate Report

Figure 3.5: Depreciation Prediction Results Output Screen

3.6 Test Cases

Table 3.3 presents the test cases used to check whether the prototype functions correctly, including valid input, prediction generation, result display, reporting, and validation scenarios.

Table 3.3: Test Cases Table

TC ID	User Story	Test Scenario	Test Steps	Expected Result
TC1	REQ-PY-1	Enter asset data	1) Open system 2) Enter cost, age, usage, maintenance, type 3) Click predict	Data is accepted and processed
TC2	REQ-PY-2	Generate prediction	1) Enter valid inputs 2) Click predict	System generates predicted asset value and depreciation
TC3	REQ-PY-3	Display results	1) Complete prediction process	Results are displayed clearly on output screen
TC4	REQ-PY-4	Maintenance estimation	1) Enter asset data 2) Run prediction	Estimated maintenance cost is displayed
TC6	REQ-PY-5	System feedback	1) Enter data 2) Run prediction	System provides feedback on input validity
TC7	REQ-PY-6	Generate report	1) View results 2) Click generate report	Report is generated successfully

TC8	REQ-PY-1	Invalid cost input	1) Enter cost = -5000 2) Fill other fields 3) Click predict	System rejects input and displays error message (cost must be positive)
TC9	REQ-PY-2	Unrealistic age input	1) Enter age = -2 years 2) Fill other fields 3) Click predict	System rejects input and displays error message (age cannot be negative)
TC10	REQ-PY-2	Invalid prediction output	1) Enter valid inputs 2) Run prediction	System detects illogical result, such as negative depreciation or value exceeding cost, and displays warning or prevents output

Chapter 4

Research Methodology

This chapter presents the research methodology used in this thesis. The methodology selected is the Design Science Research Methodology, which provides a structured process for creating and evaluating a designed solution. In this thesis, the designed solution is represented by the fixed asset depreciation prediction prototype.

4.1 Design science DSRM

The Research Methodology used in this thesis is the Design Science Research Methodology (DSRM) concept which was developed by Peffers, Tuunanen, Rothenberger, and Chatterjee (2007). It is a methodology applied to artifact development and evaluation in order to address problems with the help of information technologies. It is frequently utilized in information systems studies owing to its systematic nature and applicability in developing new solutions for problems in question.

4.1.1 Identify problem & motivate

The first step of the DSRM methodological approach is the definition of the problem to solve and the motivation of its solving. At this point, the problem in question is defined and it is stressed out why it is important to solve it.

4.1.2 Define objectives of a solution

Then the objectives of the solution are set. In other words, the primary objective at this point is the definition of the desired effect produced by the solution.

4.1.3 Design & Development

The next step will be associated with design and development of the artifact. In other words, it will be connected with creation of the artifact based on the offered solution. In most cases, it will be related to a model, prototype or some system. While design will be aimed at creation of the artifact, development will focus on proper construction of the artifact.

4.1.4 Demonstration

After the completion of the development process, it will be required to demonstrate the artifact in operation. The primary aim in such a case will be connected with showing how the artifact is capable of solving the offered problem. It can include practical demonstrations of work on some specific examples.

4.1.5 Evaluation

It will be crucial to evaluate the artifact after its development. It will help to measure the level of achievement of goals and solving of the problem.

4.1.6 Communication

Fourth, the communication phase involves the sharing of information regarding the findings from the study with different people interested in knowing about it. This includes arranging all the information regarding the problem, its possible solution, the methods used to find the solution, and the result of the exercise systematically.

Chapter 5

Results

This chapter presents the results of applying the Design Science Research Methodology to the proposed prototype, including the problem identification, solution objectives, design and development, demonstration, expert evaluation, improvements, and communication of the final artifact.

5.1 Application of Design Science Research Methodology (DSRM)

This thesis applies the Design Science Research Methodology (DSRM) proposed by Peffers et al. (2007) to guide the development and evaluation of the proposed prototype system. The methodology was selected because it provides a structured process for designing and evaluating artifacts that solve practical problems through the use of information technology. In this thesis, the artifact represents a prototype system developed to support fixed asset management through a machine learning-

based depreciation prediction concept.

5.1.1 Identify problem & motivate

The first step in the DSRM methodology deals with recognizing the problem associated with the traditional methods of depreciation accounting for fixed assets, these methods are often inflexible and do not consider additional factors such as the level of usage and the cost of maintenance so they are unable to accurately represent the actual condition and worth of the assets involved. Such problem has made the need to create a new method which would be able to incorporate modern machine learning approaches increase such as the regression machine learning algorithm proposed in this thesis.

5.1.2 Define objectives of a solution

Given the problem, the goal of the proposed solution becomes the design of a prototype model that would assist in predicting depreciation for fixed asset management. The model was intended to allow the user to enter information about the assets such as the cost of the asset, the age of the asset, whether or not the asset is under maintenance, whether the asset is being used, and the type of asset.

5.1.3 Design & Development

The Design and Development Phase was focused on developing the artifact as a prototype system using Figma in designing the prototype, an effort was made to mimic the behavior of a web application by creating two main interfaces namely

input screen displayed in Figure 3.4 and the output screen displayed in Figure 3.5 the input screen was meant to collect asset-related information and apply certain business rules on the data, while the output screen was supposed to give results such as predicted asset value, depreciation amount, and estimated maintenance cost.

Apart from creating these input/output screens other designs were also created as part of the system's design. These designs included BPMN diagrams as displayed in Figures 3.2 and 3.3, use case diagrams displayed in Figure 3.1.

5.1.4 Demonstration

The demonstration phase was aimed at presenting the prototype to the experts in both fields the technical and accounting to get their feedback on how the system works and how well it is designed to ensure that the proposed system accurately captures the intended functionalities and processes the experts reviewed the prototype screens, BPMN diagrams, use case diagrams, user stories, and test cases.

5.1.5 Evaluation

The evaluation process assessed the usability, understandability, and practical relevance of the prototype through expert feedback. Two experts reviewed the artifact: one specialized in artificial intelligence and data science, while the other specialized in accounting and fixed asset management. They evaluated the prototype screens, BPMN diagrams, use case diagram, user stories, and test cases to determine whether the system addressed the intended fixed asset management

problem.

Technical Expert: was Abdallah Eltayeb he has 7 years of experience in AI, Software Development and Technical oriented System design and currently works at DentiBit.

Accounting Expert: was Amr Essam he has 8 years of experience in accounting, involving fixed assets accounting, reporting, and review but the accounting expert's company cannot be identified for security reasons.

The technical expert said that the prototype workflow is logical and the use of regression-based ML is correct in order to predict numeric values such as depreciation amount, maintenance cost, and asset value moreover regression is sufficient to achieve the scope of prototype development while any additional progress in the field of machine learning techniques is not necessary also he suggested making several improvements in prediction BPMN which are depicted in Figure 5.2 these include adding an activity of regression-based processing and strengthening validation as well as the creation of asset insight information before showing predictions on the output screen also mentioned some improvements to the output screen shown in Figure 5.1 the output screen will be expected to display the summary of inputs, validation feedback, and asset insight and that the existing reporting BPMN shows the functionality of the reporting process required for prototype development so there is no need for making changes in BPMN lastly He also reviewed the user stories and test cases and said that some requirements overlapped around reporting and result viewing so the improved user stories shown in Table 5.1 and the improved test cases shown in Table 5.2 were refined to reduce repetition and focus on asset input, depreciation prediction, reporting, validation, and asset insights for example (REQ-PY-4) was updated to include asset insights,

(REQ-PY-5) was changed to focus on audit reporting and (TC9) was changed to test missing mandatory fields instead of invalid prediction output.

The accounting expert thought that the prototype can help in decision-making on fixed asset management because it provides useful outcomes such as depreciation prediction, maintenance cost prediction, and asset value and it also saves time compared with the more traditional ways of prediction such as straight line technique he also suggested adding an asset condition insight to the output screen which is shown in Figure 5.1 so that it would be easier for the user to understand the results in terms of asset management and accounting additionally the accounting expert evaluated the use case diagram and the user stories and the test cases and suggested refining (REQ-PY-3) so that the results view focuses on presenting prediction outputs clearly for the accountant also making sure (TC3) mentioned displayed clearly on the output screen to make sure the accounting understands the output which is shown in the improved user stories in Table 5.1 and improved test cases in Table 5.2 also based on his recommendation that the Auditor actor must be added to the use case diagram because auditing will continue to be a vital component of fixed asset management however the Auditor actor should be connected to reporting features but not prediction features also we adjusted the case diagram so it matches the user stories is shown in Figure 5.3.

Improved User Stories

Table 5.1: Improved User Stories Table

--	--

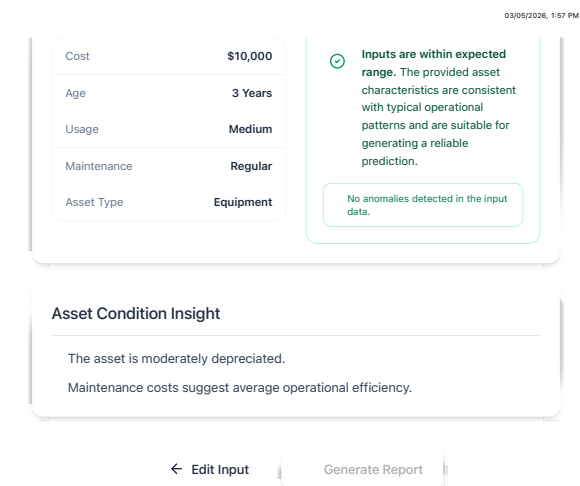
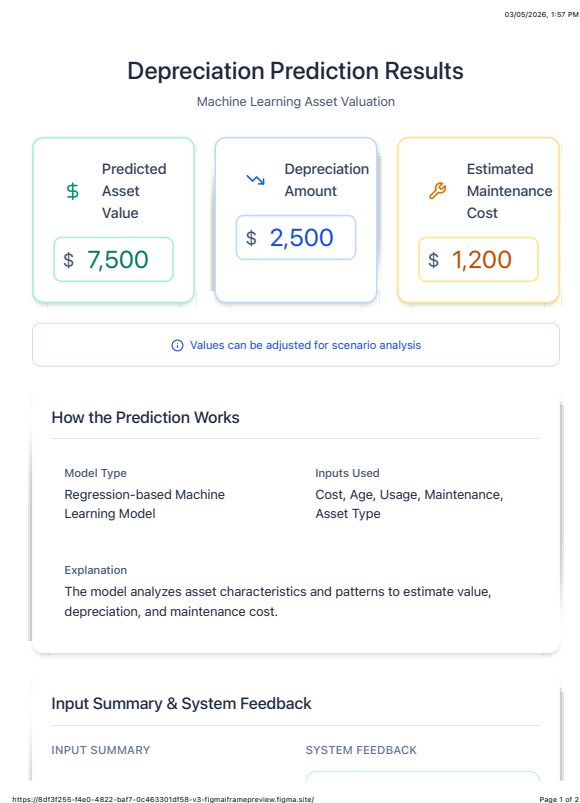
Requirement ID	Epic	User Story Summary
REQ-PY-1	Asset Input	As an Accountant, I want to enter asset data (cost, age, usage, maintenance, asset type) so that the system can process it for prediction.
REQ-PY-2	Prediction	As an Accountant, I want the system to generate predicted asset value and depreciation so that I can analyze asset performance.
REQ-PY-3	Results View	As an Accountant, I want to view the predicted results clearly so that I can understand the output.
REQ-PY-4	Maintenance Insight	As a Manager, I want to see estimated maintenance cost and asset insights so that I can evaluate asset efficiency.
REQ-PY-5	Audit Reporting	As an Auditor, I want to review the generated reports so that I can use the prediction results for auditing purposes.
REQ-PY-6	Reporting	As a Manager, I want to generate reports so that I can use the results for decision-making.

Improved Test Cases

Table 5.2: Improved Test Cases Table

TC ID	User Story	Test Scenario	Test Steps	Expected Result
TC1	REQ-PY-1	Valid asset data input	1) Enter asset cost 2) Enter age 3) Select usage and maintenance 4) Submit data	System accepts valid input data successfully
TC2	REQ-PY-2	Generate depreciation prediction	1) Enter valid asset data 2) Submit prediction request	System displays predicted depreciation amount and asset value
TC3	REQ-PY-3	Display prediction results	1) Submit valid asset data 2) Open results screen	System displays prediction outputs clearly
TC4	REQ-PY-4	Generate maintenance insight	1) Enter maintenance information 2) Submit data	System displays estimated maintenance cost and asset insights
TC5	REQ-PY-5	Generate audit report	1) Generate prediction results 2) Request report generation	System generates report for auditing purposes successfully
TC6	REQ-PY-6	Generate management report	1) Generate prediction results 2) Generate report	System generates report for decision-making purposes

TC7	REQ-PY-1	Invalid negative asset cost	1) Enter negative asset cost 2) Submit data	System displays validation error message
TC8	REQ-PY-1	Invalid negative asset age	1) Enter negative asset age 2) Submit data	System prevents prediction and displays validation feedback
TC9	REQ-PY-1	Missing mandatory fields	1) Leave required fields empty 2) Submit data	System requests completion of missing information before prediction



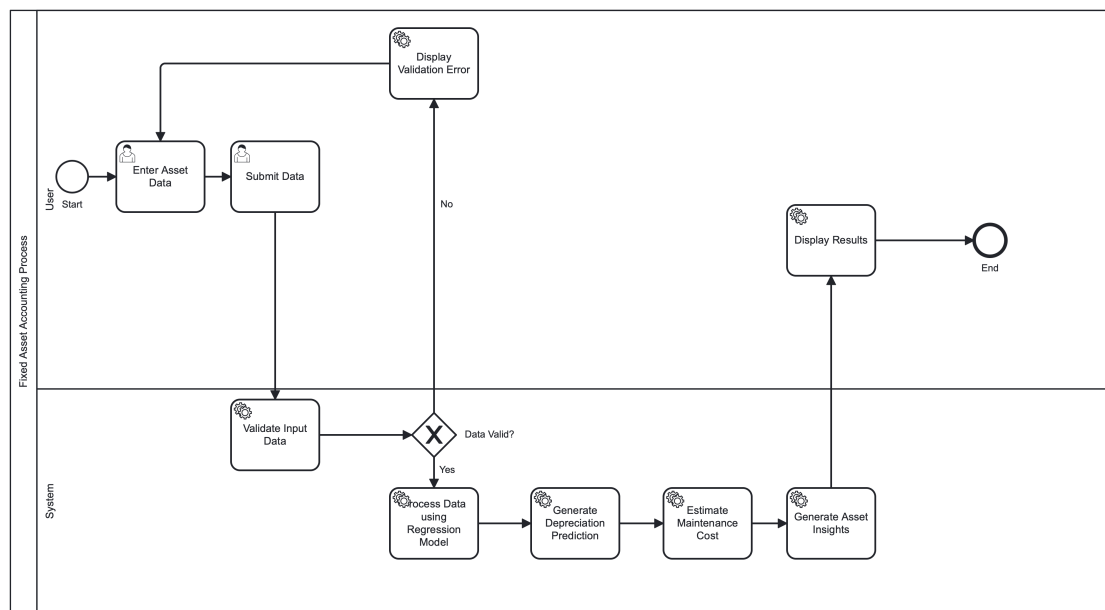


Figure 5.2: Improved Depreciation Prediction BPMN

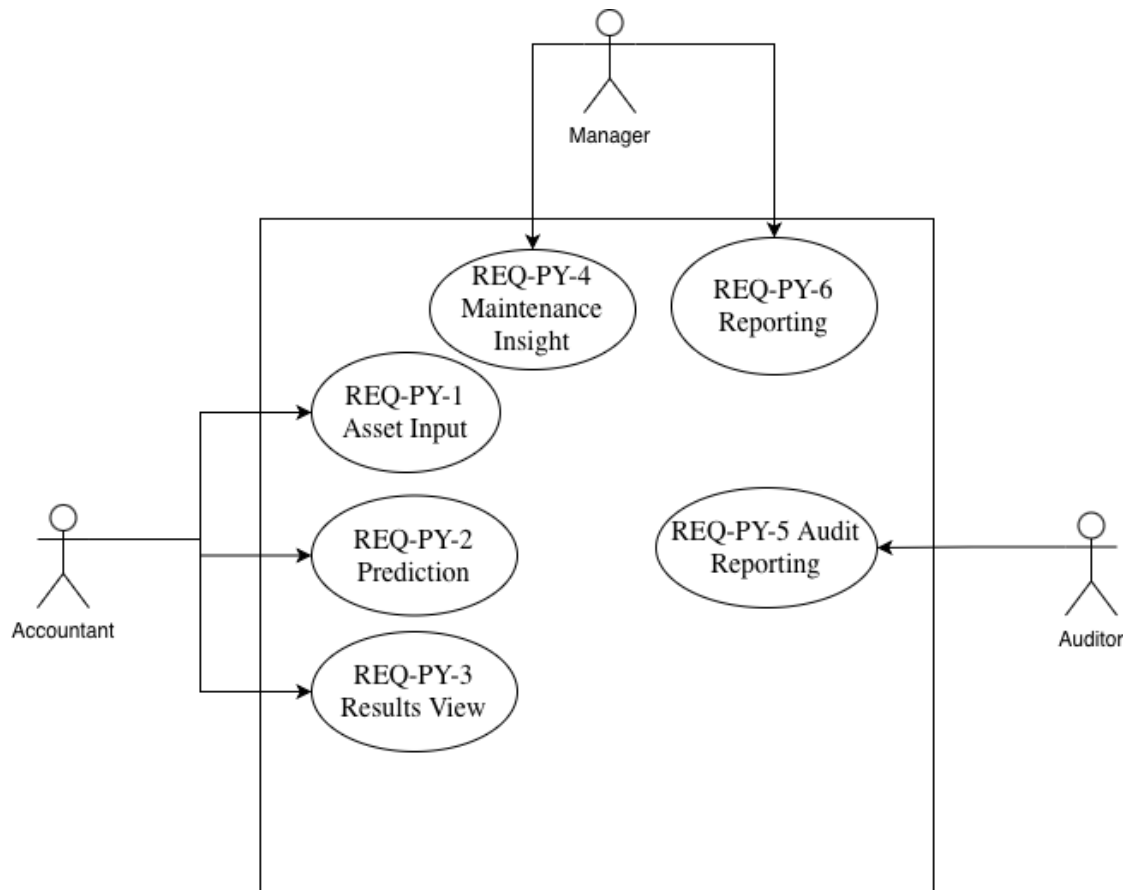


Figure 5.3: Improved Use Case Diagram

5.1.6 Communication

The communication phase is represented by this thesis itself. The main purpose of this phase is to communicate the whole research process in a clear academic form. This includes explaining the problem, the proposed solution, the design of the prototype, the expert evaluation, and the improvements made after receiving feedback.

In this thesis, the artifact is communicated through the prototype screens, BPMN diagrams, use case diagram, user stories, test cases, field requirements, business rules, interview questions, and interview responses. The thesis also presents the original design and the improved design in order to show how the artifact changed after expert feedback.

The improvement process is shown through the improved output screen in Figure 5.1, the improved prediction BPMN in Figure 5.2, and the improved use case diagram in Figure 5.3. The improved user stories shown in Table 5.1 and the improved test cases shown in Table 5.2 also communicate the final version of the prototype requirements and testing scenarios.

Therefore, the communication phase documents how the prototype was designed, evaluated, and improved throughout the thesis.

Chapter 6

Discussion

This chapter discusses the results found in Chapter 5 in this chapter i will describe feedback given by both the technical and the accounting expert what part I adjusted in the prototype and what I learned out of the process.

The first version of the prototype was about the overall idea of how the use of machine learning helps to predict the depreciation of fixed assets after having received expert feedback it was obvious that the prototype required better explanations more validation and clearer connection between the screens, BPMN diagrams, use case diagram, user stories and test cases, the technical expert validated the use of regression-based machine learning suitable for the prototype because of the fact that we predict a numerical variable (asset value, depreciation, maintenance cost)though he did suggest refining the BPMN regarding prediction so that the automation part was less confusing.

The accounting expert focused more on the business purpose and functionality firstly he recommended adding an insight into the condition of the asset so the

user can see not just the predicted value but also the interpretation behind it secondly he recommended keeping the auditor but limiting its functionality to only reports.

I think the biggest flaw within the system was that not all parts of the system were clear to each other and would be hard to use for example if I mention some functionality in my output screen i should also write some user stories for it and also have a test case for it this has been taken care of thanks to the expert feedback.

In my opinion the most valuable lesson is that technically oriented people pay more attention to the workflow and validity and model appropriateness meanwhile business users focus on clarity and usability and interpretation of results.

As for anyone designing a similar prototype i would recommended to keep the system scope clear from the beginning and make sure that the screens, BPMN diagrams, use case diagram, user stories, and test cases all match each other it's also important to collect feedback from both technical and business sides because the technical side checks the logic of the system while the business side checks if the results are useful and useable.

Chapter 7

Conclusion and Future Work

The purpose of this thesis was to illustrate how artificial intelligence may contribute in managing fixed assets, specifically via the creation of a prototype that can predict depreciation. The prototype was designed to explain how certain fixed asset data inputs such as cost, age, usage, maintenance and asset type can influence prediction outputs such as asset value, depreciation amount and maintenance cost. The design science research methodology was utilized; artifact was created, demonstrated, evaluated and enhanced with expert comments. The developed prototype contains both input and output screens, BPMN diagrams, use case diagram, user stories, test cases and business rules. It is concluded that even without the creation of a functional system a prototype can illustrate the benefits of using machine learning to aid decisions in managing fixed assets.

The current limitations of this thesis are as follows: the prototype has not been made into a functional application. Machine learning model is conceptually explained in the form of regression-based calculation, but was not actually trained

nor tested against any real data of fixed asset, as well as the fact that expert feedback was obtained from merely 2 people (one technical and one accounting professional) only and the design review was conducted through expert opinion instead of in a business environment. Therefore, all the findings in this thesis should be taken as a design study in the form of a prototype.

Future work could overcome these limitations by turning the prototype into an application running on the web and linking the application to actual machine learning model, with a real data set of fixed assets. Furthermore, expert opinions should be solicited from different disciplines, namely auditors, accountants, asset managers, IT systems developers and ERP experts. By doing so the system will not be considered from one point of view. Machine learning model could be compared by testing which model predicts the depreciation amount more precisely than others. Moreover, the prototype could be integrated to the accounting and or ERP system so that asset data could be collected, integrated and processed automatically and hence increase its value for business organizations that manage great quantity of fixed assets.

References

- Bughin, J., Hazan, E., Sree Ramaswamy, P., DC, W., Chu, M., et al. (2017). Artificial intelligence the next digital frontier. [HIGHLY CITED]. (McKinsey & Company)
- Chen, M. (2025). Agentic ai and the future of institutional asset management. *Journal of Portfolio Management*, 51(10). [HIGHLY RELEVANT]. (Journal of Portfolio Management)
- Davenport, T. H. (2018). From analytics to artificial intelligence. *Journal of Business Analytics*, 1(2), 73–80. (Taylor & Francis)
- Ford, G. L., Ancell, G., Hill, E. S., Levine, J., Reali, C., Rijks, E., & Sanchis, G. (2022). Strategic asset management. In *Power system assets: Investment, management, methods and practices* (pp. 97–140). [HIGHLY RELEVANT]. (Springer)
- Fu, Q., & Li, X. (2022). The application of artificial intelligence technology in the asset management of start-ups in the context of deep learning. *Computational Intelligence and Neuroscience*, 2022(1), 1756470. (Wiley Online Library)
- Gavrikova, E., Volkova, I., & Burda, Y. (2020). Strategic aspects of asset management: An overview of current research. *Sustainability*, 12(15), 5955. [HIGHLY RELEVANT]. (MDPI)

- Hou, B.-C., Yu, W.-T., Lu, X.-B., Yang, C.-w., et al. (2017). Applications of artificial intelligence in intelligent manufacturing: a review. *Frontiers of Information Technology & Electronic Engineering*, 18(1), 86–96. [HIGHLY CITED]. (ZUP)
- Huang, M.-H., & Rust, R. T. (2018). Artificial intelligence in service. *Journal of service research*, 21(2), 155–172. (Sage Publications Sage CA: Los Angeles, CA)
- Jiang, Y., Li, X., Luo, H., Yin, S., & Kaynak, O. (2022). Quo vadis artificial intelligence? *Discover Artificial Intelligence*, 2(1), 4. (Springer)
- Kitsios, F., & Kamariotou, M. (2021). Artificial intelligence and business strategy towards digital transformation: A research agenda. *Sustainability*, 13(4), 2025. (MDPI)
- Kogan, G., Kokina, J., Stampone, A., & Boyle, D. M. (2024). Rpa in accounting risk and internal control: Insights from rpa program managers. *Accounting Horizons*, 38(4), 137–148. [HIGHLY RELEVANT]. (American Accounting Association)
- Mogaji, E., & Nguyen, N. P. (2022). Managers’ understanding of artificial intelligence in relation to marketing financial services: insights from a cross-country study. *International Journal of Bank Marketing*, 40(6), 1272–1298. (Emerald Publishing Limited)
- Ocuppaugh, J., Roscoe, R. D., Baker, R. S., Hutt, S., & Aguilar, S. J. (2024). Toward asset-based instruction and assessment in artificial intelligence in education. *International Journal of Artificial Intelligence in Education*, 34(4), 1559–1598. (Springer)
- Paul, P. O., Ogugua, J. O., & Eyo-Udo, N. L. (2024). Innovations in fixed asset

- management: Enhancing efficiency through advanced tracking and maintenance systems. *International Journal of Science and Technology Research Archive*, 7(01), 019–026. (International Journal of Science and Technology Research Archive)
- Peppers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of management information systems*, 24(3), 45–77. (Taylor & Francis)
- Santos, P., Datia, N., Pato, M., Sobral, J., Gomes, N., Leitão, N., & Ferreira, M. R. (2023). Nlp for enterprise asset management: An emerging paradigm. In *2023 27th international conference information visualisation (iv)* (pp. 238–243). (IEEE)
- Tulebayev, E. B., Mukhanova, A. A., & Goranin, N. (2025). Predictive risk assessment of municipal assets using machine learning: modeling age, depreciation, and usage for maintenance decisions. *International journal of information and communication technologies*, 6(2), 186–199. (International Information Technology University)
- Wamba-Taguimdje, S.-L., Fosso Wamba, S., Kala Kamdjoug, J. R., & Tchatchouang Wanko, C. E. (2020). Influence of artificial intelligence (ai) on firm performance: the business value of ai-based transformation projects. *Business process management journal*, 26(7), 1893–1924. (Emerald Publishing Limited)

Appendix A

Prototype Screenshots, Figma Links, and Prompts

This appendix presents the prototype screenshots used in the thesis together with the their respective Figma links and the prompts used to generate each prototype screen.

A.1 Asset Depreciation Prediction Input Screen Prompt and Figma Link

The input screen prototype was created using Figma. The purpose of this screen is to allow users to enter fixed asset data that can be used for depreciation prediction. The Figma link and the prompt used to generate the screen are provided below.

Figma Link:

<https://www.figma.com/make/Gy6COU1dVPS9ddjf6diuYZ/Fixed-Asset-Input>

-Screen?p=f&t=hcASyYvTi1HgVMyq-0

Prompt Used:

Design a modern and clean web application input screen for a Fixed Asset Management system.

The purpose of the screen is to allow users to enter asset data in order to predict depreciation using a machine learning model.

Include the following elements:

Title at the top: "Asset Depreciation Prediction"

A simple and professional layout (centered form or card style)

Input fields:

Asset Cost (numeric input)

Asset Age (numeric input, in years)

Usage Level (numeric or dropdown: low, medium, high)

Maintenance Cost (numeric input)

Asset Type (dropdown: e.g., Machinery, Vehicle, Equipment)

Buttons:

Primary button: "Predict Depreciation"

Additional UI elements:

Small helper text under the form: "The system uses a machine learning model to estimate asset value and depreciation."

Optional icons for each input field (e.g., money, calendar, tools)

Design style:

Minimal, modern UI

Soft colors (light background, blue or green accents)

Clear spacing and readable typography

Professional look suitable for accounting/business users

Make the layout intuitive and easy to use, with proper alignment and spacing.

A.2 Depreciation Prediction Results Output Screen Prompt and Figma Link

The output screen prototype was created using Figma. The purpose of this screen is to display the results of the asset depreciation prediction generated by the machine learning concept. The screen also allows the output values to be edited for demonstration purposes. The Figma link and the prompt used to generate the screen are provided below.

Figma Link:

<https://www.figma.com/make/6ntdgHgR8iBcsd3rXgxXws/Depreciation-Prediction-Results-Screen?p=f&t=98E8wmBpo1s5fKKu-0>

Prompt Used:

Design a modern and clean web application output screen for a Fixed

Asset Management system.

The purpose of the screen is to display and allow editing of the results of an asset depreciation prediction generated by a machine learning model.

Layout:

Use a clean dashboard-style layout with a centered card or panel

Title at the top: “Depreciation Prediction Results”

Display the following outputs clearly:

1. Predicted Asset Value
2. Depreciation Amount
3. Estimated Maintenance Cost

Each output should:

Be displayed in separate cards or sections

Include clear labels and large, readable values

Be editable (use input-style fields or editable text boxes so values can be changed manually)

Include realistic example values in the design:

Predicted Asset Value: 7,500

Depreciation Amount: 2,500

Estimated Maintenance Cost: 1,200

Include an additional section titled:

“How the Prediction Works”

Inside this section, display:

Model Type: Regression-based Machine Learning Model

Inputs Used: Cost, Age, Usage, Maintenance, Asset Type

Short explanation: “The model analyzes asset characteristics and historical patterns to estimate current value, depreciation, and expected maintenance cost.”

Buttons:

“Back” or “Edit Input”

Optional: “Generate Report”

Design style:

Minimal, modern UI

Professional look for accounting/business users

Soft colors (white/gray background, blue or green accents)

Clear spacing and alignment

Dashboard-style presentation

Ensure the design is clean, easy to understand, and allows easy editing of output values for demonstration purposes in a thesis.

A.3 Improved Depreciation Prediction Results

Output Screen Prompt and Figma Link

The improved output screen prototype was created after incorporating expert feedback. The purpose of this screen is to display the depreciation prediction results more clearly and to provide additional explanatory features such as input summaries, system feedback, and asset condition insights. The Figma link and the prompt used to generate the improved screen are provided below.

Figma Link:

<https://www.figma.com/make/Ks3hPctNsbo5J3x8jyNMHi/Depreciation-Prediction-Results-Screen?t=YVzqYi8ai8rXJ2HR-0>

Prompt Used:

Design a modern and clean web application output screen for a Fixed Asset Management system.

The purpose of the screen is to display and allow editing of the results of an asset depreciation prediction generated by a machine learning model.

Layout:

Clean dashboard-style layout with a centered card or panel

Title: “Depreciation Prediction Results”

Display the following outputs:

1. Predicted Asset Value (highlighted)

2. Depreciation Amount

3. Estimated Maintenance Cost

Each output should:

Be displayed in separate cards

Include large, readable values

Be editable (input-style or editable text)

Use example values:

Predicted Asset Value: 7,500

Depreciation Amount: 2,500

Estimated Maintenance Cost: 1,200

Add a section:

“How the Prediction Works”

Include:

Model Type: Regression-based Machine Learning Model

Inputs Used: Cost, Age, Usage, Maintenance, Asset Type

Explanation: “The model analyzes asset characteristics and patterns to estimate value, depreciation, and maintenance cost.”

ADDITION 1 (Suggested by AI Expert): Include a section titled: “Input Summary & System Feedback” Display:

The input values used for prediction (cost, age, usage, maintenance, as-

set type) A short system message such as: “Inputs are within expected range”

“High maintenance cost may influence the prediction”

ADDITION 2 (Suggested by Accounting Expert): Include a section titled: “Asset Condition Insight”

Display a short interpretation: “The asset is moderately depreciated.”

“Maintenance costs suggest average operational efficiency.”

Buttons: “Back” or “Edit Input”

“Generate Report” Optional: Note: “Values can be adjusted for scenario analysis” Design style: Minimal, modern UI Professional for accounting/business users

Soft colors (white/gray background, blue/green accents)

Clean spacing and alignment

Appendix B

Interview Questions

B.1 Technical Expert Interview Questions

1. Do the proposed user stories accurately represent the intended functionalities and workflow of the system?
2. Is the selected machine learning approach, specifically regression, appropriate for the proposed depreciation prediction concept, or would another technique be more suitable?
3. Does the developed prototype include the necessary components and functionalities required to represent the proposed system effectively?
4. Are there any features or components within the prototype that appear unnecessary or not directly relevant to the system's objectives?

B.2 Accounting Expert Interview Questions

1. Do the prototype outputs support fixed asset management decision-making?
2. Are the results easy to understand from an accounting perspective?
3. Are the use cases and user stories relevant to fixed asset management?
4. Should the Auditor actor remain in the use case diagram?

Appendix C

Interview Responses

C.1 Technical Expert Interview Responses

Question 1

Do the proposed user stories accurately represent the intended functionalities and workflow of the system?

Response 1

The expert stated that the user stories mostly represent the workflow of the system. They cover the main functions such as asset input, depreciation prediction, viewing results, maintenance-related outputs, and reporting. However, he mentioned that some requirements overlapped, especially around reporting and result viewing. He suggested refining the user stories so that they focus more clearly on asset input, depreciation prediction, reporting, validation, and asset insights.

Question 2

Is the selected machine learning approach, specifically regression, appropriate for the proposed depreciation prediction concept, or would another technique be more suitable?

Response 2

The expert stated that regression is suitable for the prototype because the system predicts numerical values such as depreciation amount, maintenance cost, and asset value. He also mentioned that since the prototype is conceptual and does not include a fully trained machine learning model, regression is enough for the scope of the thesis. More advanced machine learning techniques were not required at this stage.

Question 3

Does the developed prototype include the necessary components and functionalities required to represent the proposed system effectively?

Response 3

The expert stated that the prototype includes the main components needed to represent the system, such as the input screen, output screen, BPMN diagrams, use case diagram, user stories, and test cases. However, he suggested improving the output screen by adding an input summary, validation feedback, and asset insight information. He also noted that these additions should be reflected in the user stories and test cases.

Question 4

Are there any features or components within the prototype that appear unnecessary or not directly relevant to the system's objectives?

Response 4

The expert stated that most of the prototype components are relevant, but some parts of the original workflow should be simplified. He suggested removing document-processing or generic workflow activities that do not directly support asset input, validation, regression-based prediction, and result display. He also suggested removing the invalid prediction output test case because the prototype does not implement a real machine learning model, and replacing it with a missing mandatory fields validation case.

C.2 Accounting Expert Interview Responses

Question 1

Do the prototype outputs support fixed asset management decision-making?

Response 1

The expert stated that the prototype outputs can support fixed asset management decision-making because they provide useful information such as predicted asset value, depreciation amount, and maintenance cost prediction. He explained that these outputs can help users understand the financial and operational condition of fixed assets. He also mentioned that the prototype gives a broader view than

traditional depreciation methods because it considers factors such as age, usage, maintenance, and asset type.

Question 2

Are the results easy to understand from an accounting perspective?

Response 2

The expert stated that the results are understandable, but they would be more useful if the system added a short explanation of the asset's condition. He explained that the output screen should not only show numbers, but should also explain what these numbers mean from an asset management perspective. Based on this feedback, the asset condition insight section was added to the improved output screen.

Question 3

Are the use cases and user stories relevant to fixed asset management?

Response 3

The expert stated that the use cases and user stories are relevant because they cover asset input, prediction, result viewing, maintenance insight, and reporting. However, he suggested refining REQ-PY-3 so that the results view focuses more clearly on presenting prediction outputs for the accountant. He also suggested making sure that TC3 tests whether the prediction results are displayed clearly on the output screen.

Question 4

Should the Auditor actor remain in the use case diagram?

Response 4

The expert stated that the Auditor actor should remain in the use case diagram because auditing is still an important part of fixed asset management. However, he explained that the Auditor should not be involved in the prediction process directly. The Auditor role should be limited to report-related functions, such as reviewing or generating reports, because auditors usually verify outputs rather than enter asset data or generate predictions themselves.

Declaration

I herewith declare that this report is in full accordance with the Plagiarism Guidelines of the Faculty of Management Technology at the GUC.

Signature: Hassan Ahmed Eltayeb

.....